AIAA '89

AIAA 89-0154
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27th Aerospace Sciences Meeting
January 9–12, 1989/Reno, Nevada

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ON THE LEAN BLOW-OUT LIMIT OF PREMIXED FLAMES

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Abstract

The lean flammability limit of an air/propane premixed flame was studied experimentally, using Schlieren photography synchronized with OH-imaging done with the Planar Laser Induced Fluorescence (PLIF) technique. The flame was studied in a large range of equivalence ratios. Various steps in the process of the flame destabilization were investigated, including partial lift-off, stable lift-off, and final blow-out conditions. The flame structure was visualized for each stage showing the transition from a flame held at the nozzle to a flame held by the flow structures. In order to study the later condition in more detail the flame was forced at the preferred mode frequency generating large, stable, coherent structures in the core region. The changes in the mean and turbulent flow fields as a result of this forcing were studied, and the modified flame structure was visualized to understand the interaction between the flame and vortical flow dynamics.

Introduction

Design trends of modern airbreathing propulsion systems, such as gas-turbines or ramjets, include optimized airblast atomization systems with enhanced mixing, requirements for low idle power for reduced emission, design of a "stoichiometric combustor" for high overall fuel to air ratio with improved fuel consumption, reduced smoke limits for pollution control, and a general quest for expanded operating envelope. All these trends push the engine to operate in marginal lean stability limits. The problem of combustion of extremely lean mixtures and related problems of flammability of diffusion and premixed flames were discussed extensively in the literature.[1-5] There were also studies of methods to increase flame stability, such as using swirl,[6] for example.

Various theories have been proposed to explain flame extinction, such as effects of convection, chemical kinetics, flame stretch, diffusion of reactants, and instabilities. All these theories are somehow related to thermal balance in the flame between the heat release rate and the heat loss rate. The burning velocity, flame thickness, and flow dynamics seem to be the most fundamental parameters governing the complicated processes in the flame.

At the Naval Weapons Center (NWC) we have concentrated on the interaction between flow dynamics and the flame structure. Using the Planar Laser Induced Fluorescence (PLIF) technique we studied diffusion and premixed flames.[7-11] When the flame was forced at the preferred mode frequency of the jet,[12] or its harmonics, the reaction occurred inside vortical coherent structures.[9] In the region between the vortices, the so-called "braids" , the reaction was mostly quenched. Similar observations were reported by Lewis et al (1987)[13] and simulated numerically by Givi et al (1986).[14] This quenching was attributed to the high strain rates in the flow at these locations, associated with high velocity gradients. The flame temperature drops locally and the reaction rate reduces to zero. When local quenching occurs in the braid region between the burner and the first vortical structure it may lead to flame lift-off.

The objective of the present work is to study the flow conditions leading to flame lift-off and blow-out and to control the fluid dynamics to extend the lean blow-out limit.

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Experimental Arrangement

A system for PLIF imaging of OH radicals in flames using a XeCl excimer laser at 308 nm was used (Fig. 1).[8-11] The laser beam was expanded into a planar sheet, passed through the flame in a chosen direction: parallel to the flame axis for studying the streamwise evolution of the structures or perpendicular to the axis for azimuthal structures measurements. The resonance fluorescence from OH was imaged with a gated intensified diode array camera. The sheet was approximately 0.5 mm wide. Since only structure was of interest, no attempt was made to calibrate for absolute sensitivity and quenching for use in absolute number density calculations. Natural flame emission was removed from the images by gating the intensifier on for only a short period encompassing the laser pulse (about 100 nsec).

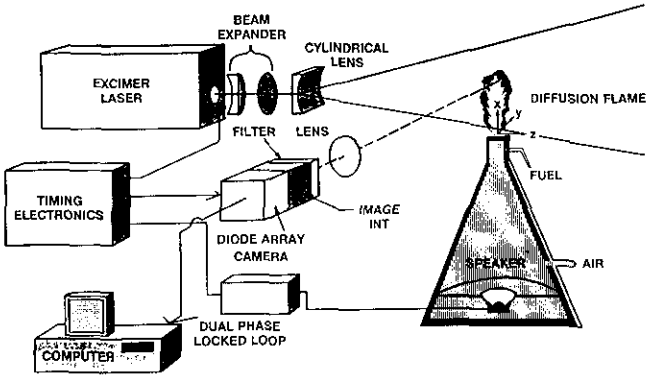

Figure 1. Experimental Set-Up.

A propane/air premixed circular flame was studied. The air was supplied through a conical settling chamber into a 210 mm long circular pipe (50 mm diameter). An orifice plate with a 19 mm diameter circular hole was mounted on the exit of the pipe. The propane was injected at the entrance to the pipe through multiple circumferential holes, normal to the pipe walls. The propane fuel flow rate varied in the tests between 0.5 to 2 l/min. The air flow rate varied in the tests from 20 to 55 l/min yielding exit velocities of 1.65 to 4.55 m/sec. The corresponding Reynolds number range was 2100 to 5800, based on the exit diameter, exit air velocity and the kinematic viscosity of the air at room temperature. The initial velocity RMS level measured with a hot wire anemometer at the exit of the nozzle on the centerline was 2.7%.

The flame was excited by a set of four speakers mounted in an acoustic resonating chamber (a 45 cm diameter base by 66 cm tall cone) which was used as a settling chamber as well. The speakers were driven at controlled frequencies and amplitudes using a dual phase locked loop and audio power amplifier. To construct the turbulent structure time history, the laser system was phase locked to acoustic excitation of the flame jet. This allowed the time history of the vortex roll-up to be obtained in multiple frames in a process similar to stroboscopic measurement of periodic machinery. The forcing frequencies, amplitudes, and relative phase angles were monitored (in cold flow) by using a calibrated hot-wire anemometer which was also used to measure the mean and turbulent velocities of the air. The speakers force, and phase lock, only the longitudinal mode of the jet. The maximal forcing level used was 15% RMS of the exit mean velocity.

The data was taken in two different modes. Short time exposures were taken yielding instantaneous OH LIF signal level images (18 nsec). Alternately, multiple frames (about 100), taken at a constant phase angle, were averaged together for each image to reduce the chaotic nature of the flow and bring out the coherent portion.

A Schlieren visualization system was mounted at 30° angle to the PLIF laser sheet. The light source for the Schlieren system (A Xe flash lamp) was triggered together with the laser to give simultaneous PLIF and Schlieren images. The Schlieren set-up included two 203 mm parabolic F10 mirrors and a 500 mm focal length, 50 mm, imaging lens. The flash lamp light duration was 0.5 μsec.

Results and Discussion

Lean Stability Limit

The lean stability limit of the flame was studied for a fuel flow rate from 0.5 l/min to 2 l/min. The fuel flow rate was kept constant, at different values in this range, and the air flow rate was increased gradually. Close to the lean limit, several changes in the flame behavior occurred before the final blow out. These changes are summarized in Figure 2. The first sign of the approaching limit was an initial circumferential instability of the flame, which started to separate from the flameholder, intermittently, at random locations around the nozzle. For slightly higher air velocity, partial lift-off was observed. Large sections of the flame would lift-off randomly around the nozzle and reattach. A stable lift-off followed for still higher air flow rates and then blow-out. The various stages of flame instability occurred across a wider range of air flow rates for the higher propane flow rates. At lower propane flow rates the blow out occurred without passing through the earlier stages. Figure 2 also shows the stoichiometric line. For propane flow rates lower than 1 l/min the flammability limit is at the air-rich side of this line. For higher fuel flow rates the flammability limit crosses the fuel-rich side of the stoichiometric line. As the air velocity increases, more air is entrained to the flame downstream of the exit, and the actual air to fuel ratio is increased.

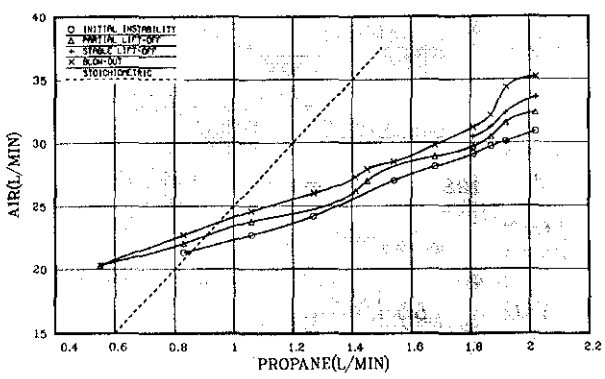

Figure 2. Flame Behavior at the Lean Flammability Limit for Various Air to Fuel Ratios.

The instantaneous structure of the flame at the different stages of instability were visualized using simultaneous Schlieren photography and OH imaging using the PLIF technique. The OH images visualize the combustion zones in the flame. Figure 3 shows a pair of Schlieren pictures and an OH image at the same conditions (air flow rate = 29.3 l/mm and propane flow rate = 1.86 l/min). The Schlieren pictures cover a much larger field of view relative to the OH image. The extent of the latter is marked on the figure with a dashed square. Figure 3a shows the Schlieren picture of the nonreacting air/propane jet before igniting the flame. The initially laminar jet develops large-scale structures in the shear layer and eventually transitions to fine-scale turbulence. The Schlieren picture of the flame for the same conditions is shown in Figure 3b. The line-of-sight integration, inherent to the Schlieren image, limits the ability to study the inner flame structure. The flame laminar core is evident and so are the large buoyancy driven structures at the edges.[15] The OH-image in Figure 3c shows the stable narrow combustion zones at the flame circumference. The flameholder (nozzle exit) is located at $x/D = 0$ in the figure. As the air flow rate is slightly increased to 31.25 l/min, the flame is partially lifted off. This condition is visualized in Figure 4 with a Schlieren (Figure 4a) and OH-image (Figure 4b). The Schlieren picture shows the flame attached to the flameholder only on one side (right). The flame's core becomes more wrinkled. Similar features can be seen in the OH image (Figure 4b). The flame is attached at the left side of the figure but lifted on the right side. The lifted flame is more turbulent than the attached flame. In the stable lift-off conditions (air flow rate = 32.1 l/mm) the flame is fully separated from the flameholder (Figure 5). The Schlieren picture (Figure 5a) shows the mixed air/propane nonreacting jet between the nozzle and the flame. The flame core is highly turbulent, masked by the large laminar buoyant structures. The corresponding OH image in Figure 5b shows the fully lifted flame,

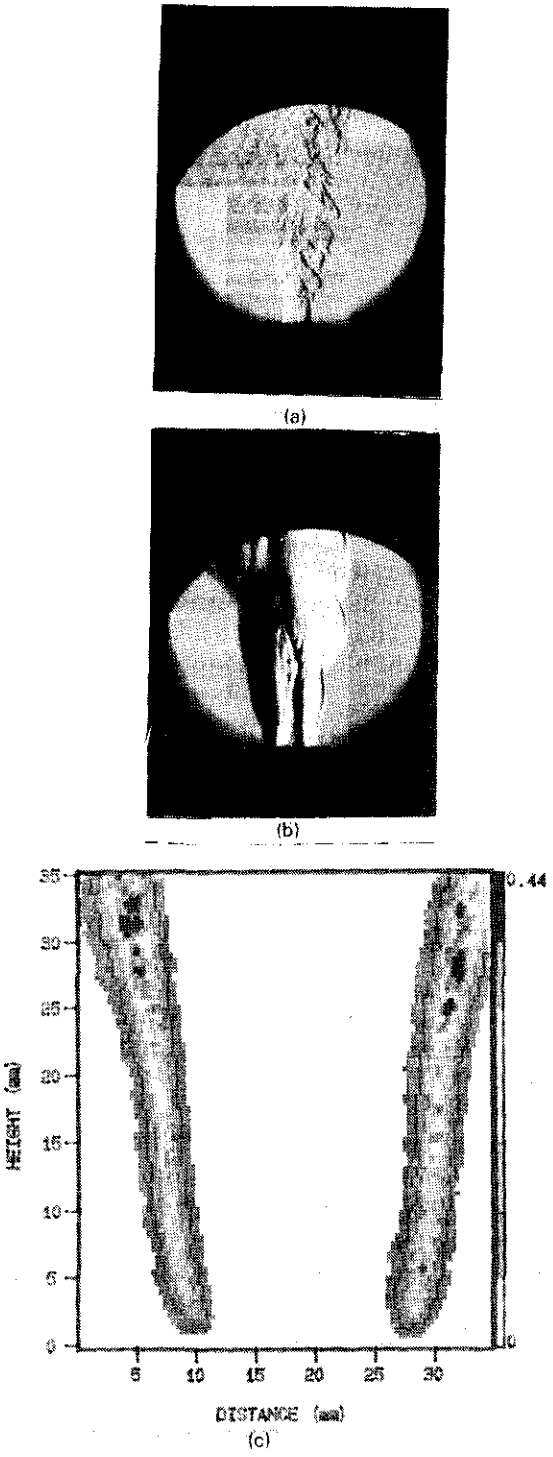

Figure 3. Instantaneous Structure of the Stable Flame Visualized by Schlieren of the Nonreacting Air/Propane Jet (a), Schlieren of the Flame (b), and OH Imaging (c). Air flow rate = 29.3 l/min; Propane flow rate = 1.86 l/min.

standing close to x/D = 0.5. The right side of the flame seems to be laminar in this picture, while the left side is held by a vortical structure. A slight increase of the air velocity(flow rate of 33 l/min) brings the flame close to the blow-out limit. This condition is depicted in Figure 6. Two Schlieren pictures are shown. Figure 6a shows the nonreacting air/propane jet and Figure 6b shows the flame. There is no discernible difference between the structure of the jet at this condition and that observed in Figure 3a for the stable flame. The flame structure is similar to the lifted flame in Figure 5a.

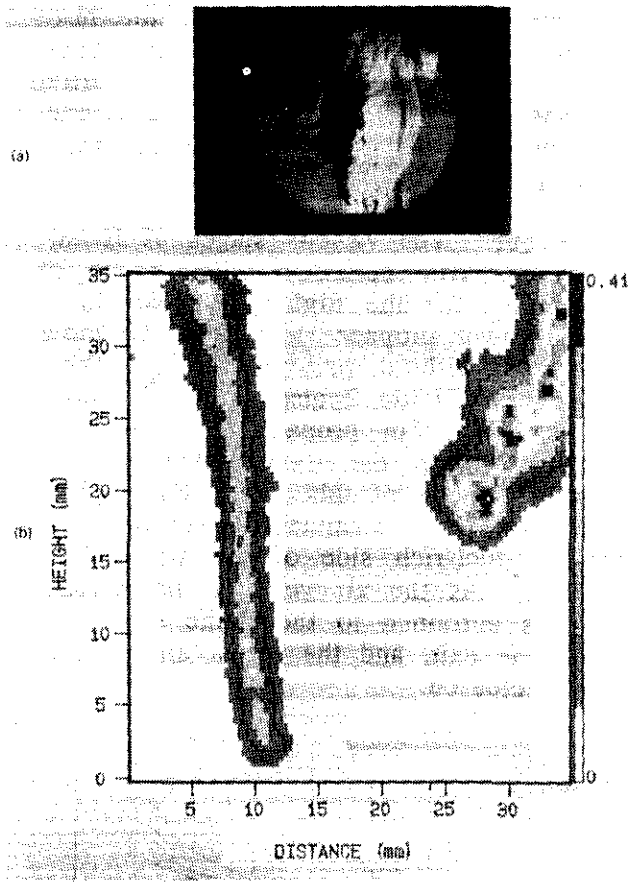

Figure 4. Instantaneous Structure of a Partially Lifted-Off Flame Visualized by Schlieren of the Flame (a) and OH Imaging (b) Air flow rate = 31.25 l/min; Propane flow rate = 1.86 l/min.

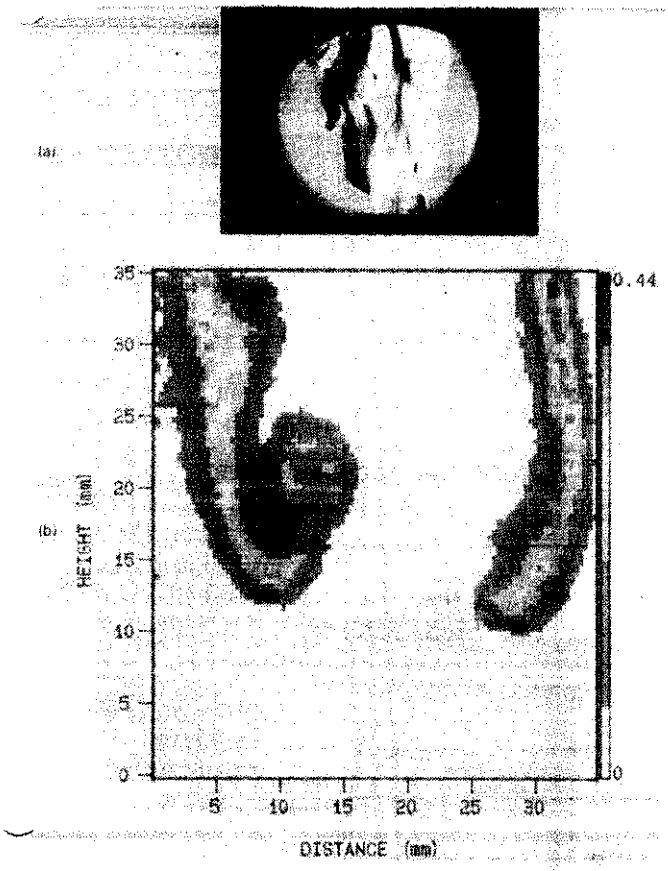

Figure 5. Instantaneous Structure of a Stable Lifted-Off Flame Visualized by Schlieren of the Flame (a) and OH Imaging (b). Air flow rate = 32.1 l/min; Propane flow rate = 1.86 l/min.

The OH image in Figure 6c shows more clearly that the lift-off height increased to x/D ≈ 1. Both sides of the flame seem to hold around vortical structures in the jet. The location of the bottom of the flame is near the location where large-scale vortices develop in the nonreacting jet (Figure 6a).

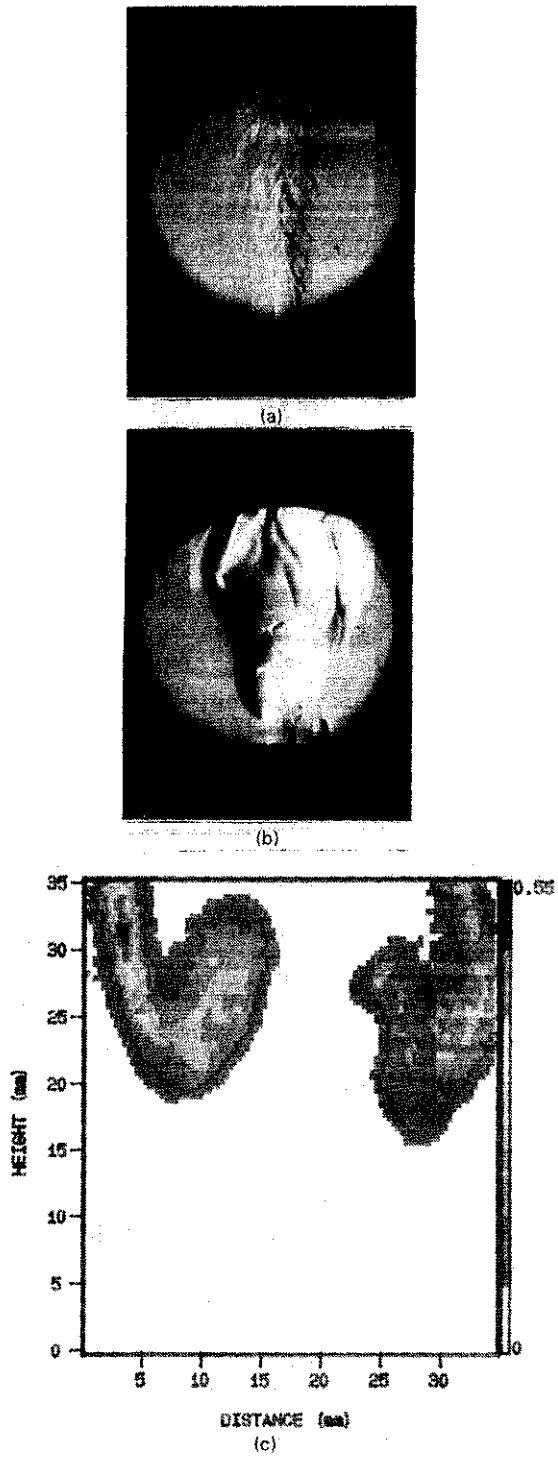

Figure 6. Instantaneous Structure of a Flame Near Blow-Out Visualized by Schlieren of the Nonreacting Air/Propane Jet (a), Schlieren of the Flame (b), and OH Imaging (c). Air flow rate = 33 l/min; Propane flow rate = 1.86 l/min.

Effect of Forcing on the Flame Structure and Stability

The stability characteristics of the flame could be related to the air/propane shear layer structures, which are governed by the nonreacting premixed jet instability. The frequency at which the jet is most responsive to forcing is the preferred mode frequency.[12] When normalized by the jet exit diameter (D) and velocity (U_0) it yields a Strouhal number, St, in the range of 0.2 to 0.5. For orifice jets the Strouhal number is even higher due to the flow acceleration downstream from the nozzle. The preferred mode frequency of the present jet was measured on the jet centerline at x/D = 2 for a wide range of Reynolds numbers ($Re = U_0 D/\upsilon$, where υ is the air kinematic viscosity). The preferred mode frequency was determined from the broad band peak in the velocity power spectrum as measured by a hot-wire anemometer. The results, given in Figure 7 showed that the preferred mode Strouhal number for this orifice jet varied between 0.47 to 0.64.

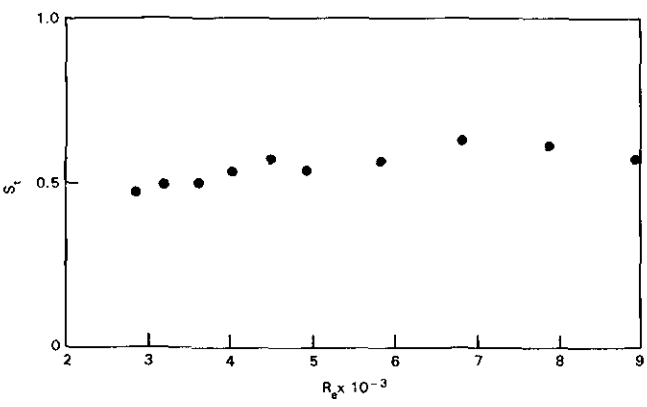

Figure 7. Variation of the Preferred Mode Strouhal Number With the Jet Reynolds Number.

Forcing the jet at the preferred mode frequency had a substantial effect on the mean velocity and the turbulence intensity. Figures 8 and 9 show the change in the mean and turbulent velocity radial profiles, respectively, as measured at the jet exit (x/D = 0.13), for two forcing levels: 5% and 15% (turbulent velocity only for 15%). The forced profiles are compared with the unforced case. The forcing causes the jet to spread significantly faster. The initial momentum thickness, θ, defined as:

$$\theta = \int_0^\infty \frac{U}{U_0}\left(1 - \frac{U}{U_0}\, dr\right), \qquad [1]$$

which is a measure of the jet width, changes from an unforced value of 0.23 mm to 0.76 mm for 5% forcing level, and to 1.68 mm for 15% forcing. The turbulence maximum intensity was only slightly changed, from 45% of the mean velocity to 50% for the higher forcing level. But along with the increase in the jet shear layer thickness, the region with high turbulence intensity became much larger. Also the turbulence intensity at the jet centerline increased from 2.9% for the unforced case to 15.6% for the high forcing.

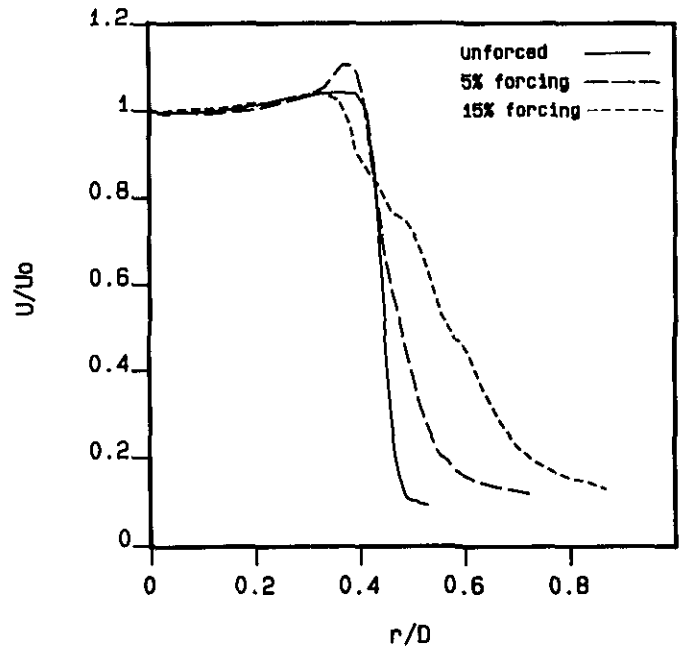

Figure 8. Mean Velocity Radial Profiles at x/D = 0.13 for Forcing at the Preferred Mode Strouhal Number, at Three Forcing Levels.

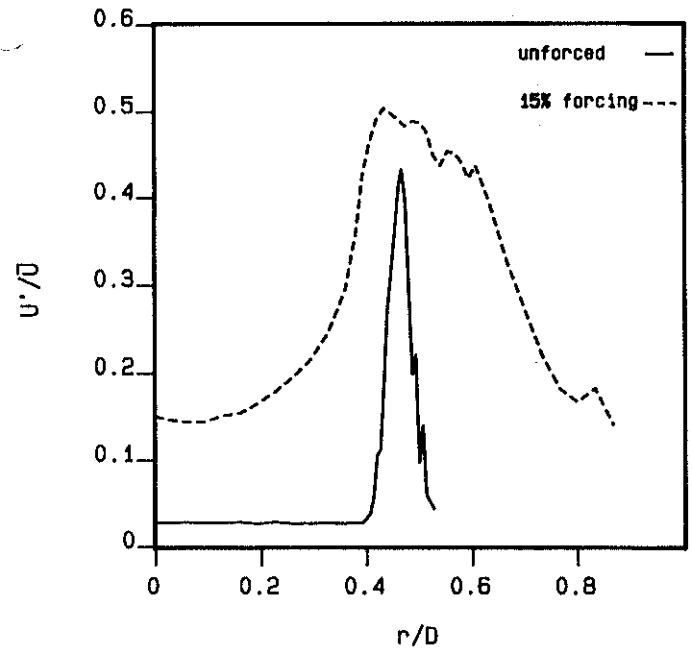

Figure 9. Turbulence Intensity Radial Profiles at x/D = 0.13 for Forcing at the Preferred Mode Strouhal Number, at Two Forcing Levels.

The variation of the jet momentum thickness with the axial distance from the nozzle is given in Figure 10. A sudden change in the slope of this graph can be discerned at x/D = 0.75. This location corresponds to the axial distance where transition was observed in the Schlieren picture of the nonreacting jet in Figure 3a and the flame is held when it is lifted off (Figure 5). It is suggested that the change in the radial velocity gradients, and the roll-up of vortical structures, enable temporary stabilization of the flame at this location, when it is lifted off the flameholder.

The large increase in the jet spreading rate is due to the generation of large-scale structures in the shear layer. These structures are visible in the Schlieren picture of the nonreacting jet in Figure 11a. Similar structures also appear in the Schlieren picture of the flame in the core region (Figure 11b).

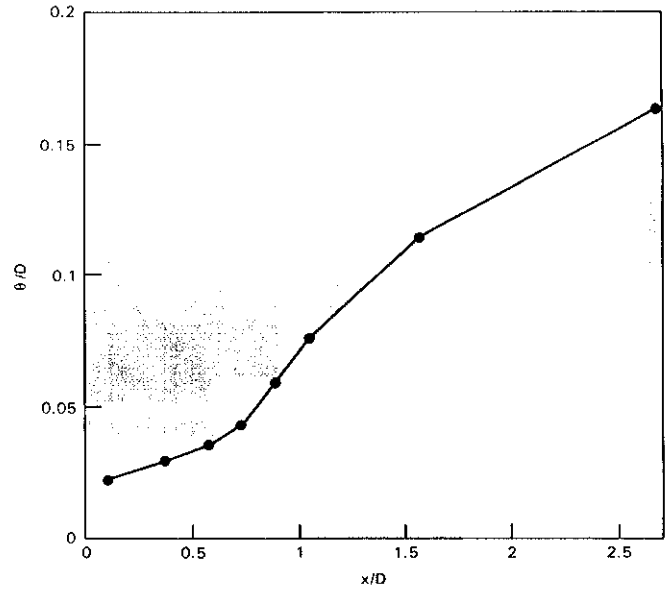

Figure 10. Vatiation of Jet Shear-Layer Momentum Thickness with Axial Distance for a Nonreacting Jet. Air flow rate = 34 l/min.

The OH-images corresponding to the same conditions are shown in Figure 12. Figure 12a depicts the phase-averaged OH concentration, while 11b gives the instantaneous (18 nsec exposure) image. The phase-averaged picture (Figure 12a) shows that the flame zone is most of the time outside the large-scale vortices, having the shape of the outer contour of these vortices. The instantaneous image (Figure 12b) shows that occasionally the flame zone penetrates the vortices and wraps around the flow structures. This behavior can be attributed to the fact that the flame studied here has an air/fuel mixture ratio slightly above the stoichiometric ratio. The right mixture ratio is obtained at the jet edges after mixing with the entrained ambient air. Since the entrained air is engulfed by the vortices, the stoichiometric ratio can also be reached inside the vortices allowing the flame to propagate into the vortices. Local quenching can occur in the braid region between the vortices, as a result of the local high strain rates, as observed in Figure 12b and discussed in Gutmark et al (1987).[9]

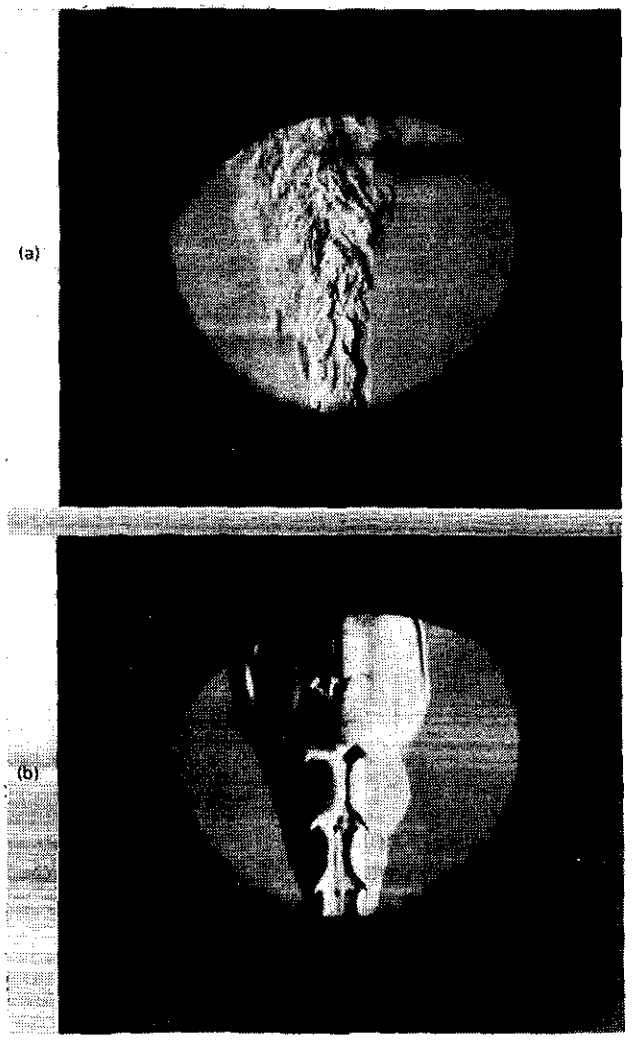

Figure 11. Schlieren Pictures of the Nonreacting (a), and Flame (b), Forced at the Preferred Mode Strouhal Number (St = 0.57), Forcing Level 13%. Air flow rate = 28.9 l/min; Propane flow rate = 1.86 l/min.

Discussion and Conclusions

The flame destabilization at the lean flammability limit occurs gradually starting from local separation of the flame sheet from the flameholder, continuing with partial lift-off for higher air velocities, followed by stable lift-off and finally complete blow-out. These processes occur over a wide range of velocities in the higher equivalence ratio conditions, while for low equivalence ratios the transition is more rapid.

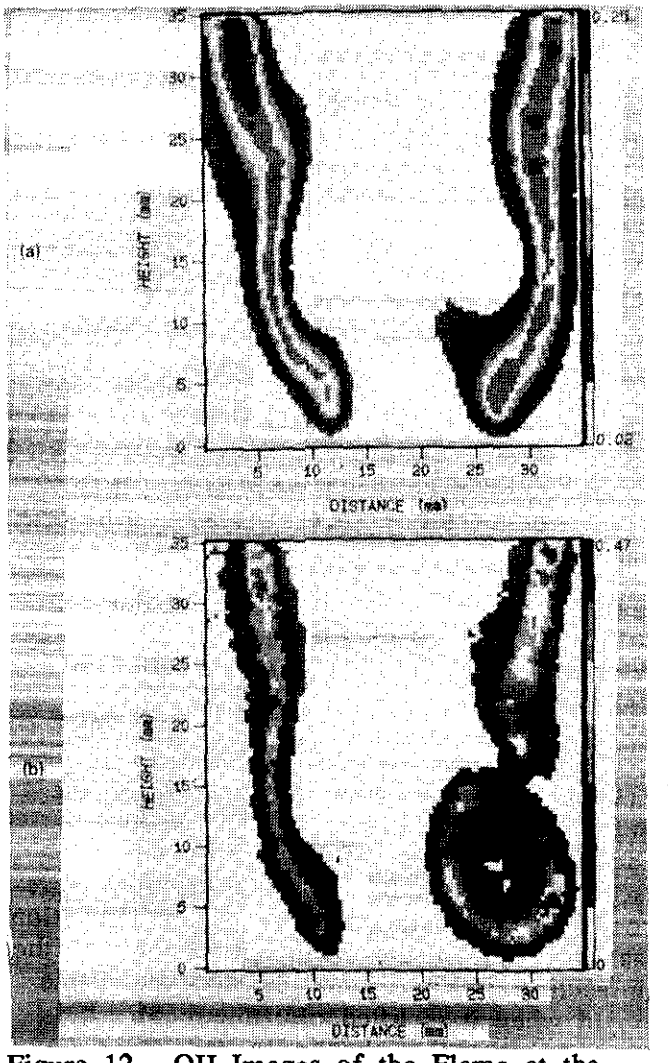

Figure 12. OH Images of the Flame at the Conditions Described in Figure 11. (a) Phase-averaged, (b) Instantaneous.

The initially laminar shear layer of the premixed air/fuel jet is very thin with high gradients of mean velocity, leading to high strain rates of the flow. The laminar shear layer transitions to turbulent state, through amplification of disturbances, leading to vortex roll-up, which eventually breakdown to small scale turbulence. For high jet velocity, when the flame can no longer be held at the flameholder, a new equilibrium point for flameholding is needed. Due too high strain rates at the laminar shear layer, the flame cannot be held there. The roll-up of the shear layer to vortices, generates region of flow reversal and stagnation

points, followed by lower velocity gradients and strain rates in the turbulent region. At this transition point the flame can be held, while there is enough unburned fuel, for premixed conditions above stoichiometric ratio.

Forcing the premixed flame at the preferred mode frequency generates large-scale structures in the flame, starting close to the flameholder surface. The roll-up of vortices generates, as discussed above, regions of flow reversals and stagnation points, where the flame can penetrate the flow and be held. But also, regions between vortices and outside the large structures, have high strain-rates that can lead to local quenching of the flame. This combination causes unsteadiness in the flame characteristics, when the flame is seen to be held alternatingly at different locations between the vortical structures.

The effect of these structures on stabilization and destabilization of the flame at the lean blow-out limit is currently being studied in our laboratories.

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